MUHAMMAD ARSALAN AKRAM

Telecom & Network Professional

Riyadh, Saudi Arabia | +966 54 198 6741 | engineer_arsalan@hotmail.com

PROFESSIONAL PROFILE

Senior Optical Network / Telecom Professional with 15+ years of experience in network design, engineering planning, deployment & commissioning, integration and operation & maintenance. Extensive hands-on experience with Nokia, Infinera, Huawei, NSN, Ciena optical platforms across Metro, Long-Haul and submarine networks. Proven experience in leading technical teams, coordinating field/NOC/customer/vendor activities, supporting product evaluation, procurement, costing and commercial activities, and improving network quality, reliability and efficiency. Holds multiple internationally recognized technical and management certifications and open for relocation.

CORE TECHNICAL SKILLS

Optical: DWDM | OTN | SDH | ASON | FTTH | FTTB | FTTX | FTTS | GPON
Submarine Networks: SMW systems | Transworld | SLTE | PFE | Repeaters | IPLC
TAC & NOC Operations: 24/7 NOC Operations and Troubleshooting | Service Assurance | SLA/KPI Monitoring | Major Incident Management | Escalation Management | Fault Management | RCA | Incident Restoration | Problem Management | Change Management | Operational Governance
Network Management & Operations: Nokia NSP | NFM-T | Wavesuite NOC | Ciena MCP (Manage, Control and Plan) | Infinera Transcend | Huawei iMaster NCE | NSN TNMS | Web LCT | DCN | Putty | WebUI | Ericson ARCGIS
Networking: IP/MPLS | BGP | OSPF | IS-IS | Layer 1/2 Troubleshooting | Juniper & Huawei Routers/Switches | Firewalls | IP Transit | IPLC | DPLC | Data Center Interconnect (DCI) | ODF/MMR Connectivity
Testing: OTDR | Insertion Loss | ORL | Optical Power | Receiver Sensitivity | OSNR | Chromatic Dispersion | PMD | Fiber/Link Quality Assessment | PAT/ATP | Analyzer (OSA) | BERT Meter | Ethernet testers | EXFO
Automation & Scripting: Python | CLI Automation | Configuration Automation | Test Automation | Configuration Validation | Network Data Processing | Deployment Workflows
Linux/Unix environments: CLI-based network operations & Troubleshooting | Configuration and scripting workflows | Network tools and utilities
Project & Technical Delivery: Network Design | Deployment | Integration | Commissioning | MOPs | Technical Documentation | Test Procedures | Acceptance Testing | Punch Lists | NCR/Non-Conformity Management | Corrective Actions | Vendor Management | Product Evaluation | Risk Management

PROFESSIONAL EXPERIENCE

Nokia – Saudi Arabia | Senior Optical Engineer / Team Lead | May 2019 – Present

- Lead end-to-end delivery of DWDM//OTN optical projects for major customers including Saudi Electricity Company / National Grid and Saudi Telecom Company (STC), covering design, planning, deployment, integration, commissioning and operations & maintenance.
- Lead design team for signal flow, HLD/LLD, SOW, MOP and BOQ using Visio, Autocad, MS word etc
- Hands-on experience across the Nokia 1830 PSS optical platform family, including PSS-32/16/8/4, PSS-24x/16x/8x/64, PSS-ONE, associated WDM/OTN cards and network elements, covering configuration, service provisioning, protection, upgrades, troubleshooting and network expansion.
- Apply optical transport experience with Infinera DTN-X, GX Series, XTM, and Transcend technologies to support engineering, platform operations, service assurance, troubleshooting, and migration activities.
- Perform optical engineering and planning using EPT (Engineering Planning Tool), supporting network architecture, capacity expansion, connectivity, signal-flow and implementation design.
- Drive optical scaling activities including fiber/link quality assessment, capacity planning, optical cutsheets, configurations and deployment documentation (MOP).
- Implement network upgradation using Nokia CLI and automation/scripting-assisted processes, improving configuration consistency and deployment efficiency.
- Develop and use automation workflows for configuration preparation, validation, network upgrade activities and operational processes.

- Manage optical network visibility, monitoring and operation using Nokia Eco system, Infinera TNMS.
- Coordinate field, NOC, engineering, customer and vendor teams to deliver rollout activities within project schedules, technical requirements and operational standards.
- Lead complex optical troubleshooting, RCA and provide technical direction for network-impacting issues.
- Support technical product evaluation, vendor coordination, procurement, costing and commercial activities.
- Provide technical leadership and mentoring, supporting decision-making on technical solutions, resources, risks, timelines and project costs.
- Supervise and coordinate field/vendor implementation activities including equipment installation, fibre connectivity, ODF/interface work, labelling and deployment documentation.
- Review implementation readiness, MOPs, installation records and acceptance evidence; track punch-list items and coordinate closure of technical deviations and corrective actions.
- Provide technical assistance to create, monitor, prioritize and restore tickets using designated tools.
- Lead service support for any incident, and conduct deep testing/troubleshooting in coordination with different team's/stake holders/customer etc.
- Manage customer network and its software like ARCGIS for NE tool, FUP, NFTS, TFS, JIRA, HUB etc.

Transworld – Pakistan | Subsea NOC Engineer, NMS/Application Lead | Jun 2015 – May 2019

- Supported 24/7 operations of large-scale DWDM, OTN, SDH, GPON, FTTx and IP/MPLS networks.
- Led TAC (Technical assistance center) department for deep troubleshooting related to submarine networks Ciena, Alcatel lucent and Huawei.
- Supported international submarine and terrestrial connectivity while maintaining service availability and SLA/KPI compliance and coordinated incident resolution to minimize service disruption and improve MTTR.
- Managed optical monitoring, service provisioning, performance analysis, fault management and restoration using vendor management systems and supported NMS and application administration, operational reporting, incident tracking and service assurance, strengthening operational governance and visibility of network performance.
- Operated multi-vendor optical platforms including Huawei OptiX 9800/8800/7500/3500/1500, Metro 5000/1000/500, NSN HiT 7065, Subcom DWDM, 1520LM Alcatel lucent and Huahuan.
- Worked with Ciena optical transport technologies, including the Ciena 6500 platform, WaveLogic Ai/5e (WL5e), Reconfigurable Line System (RLS), MCP, and SAOS.
- Led fault isolation, root-cause analysis, escalation management and service restoration, improving major-incident response, reducing recurring faults and supporting customer satisfaction.
- Supervised operational tasks related to landing station including racks, power, fibre routing, ODF/MMR termination, labelling, and documentation.
- Supported optical network planning, implementation, configurations and capacity expansion activities.
- Coordinated international bandwidth and submarine cable services with carriers and vendors for provisioning, troubleshooting and restoration.
- Supported IP/MPLS network operations using Juniper MX-series and Huawei routers and switches for IP transit, IPLC, DPLC and gateway services.
- Supported data-center operations (DCI) and network optimization.
- Supported SEA-ME-WE 5 100G submarine cable network and 100G TW1 submarine system upgrade projects.
- Operated and maintained submarine optical systems including Huawei Marine, SMW5, Alcatel, Ciena, TE SubCom SLTE, PFE, LME and optical repeaters.
- Coordinated with International BW vendors like SEA-ME-WE 3, SEA-ME-WE 4, IME-WE, SMW, SEA-ME-WE 5, TATA, Epsilon, Cogent, Google Telecom Italia Sparkle, SingTel, DU, ITC-KSA, Etisalat and Omantel etc. for creation, troubleshooting and restoration of links.

LCC International – Pakistan | Fixed Network Engineer / Team Lead | Feb 2012 – Jul 2015

- Led Level II Transmission and Access Network engineering across Huawei, Ericsson and ZTE environments.
- Delivered DWDM, OTN, GPON/FTTH, MSAN, DSLAM and IP migration projects.
- Supported optical design, link budgeting, configuration, installation, commissioning and acceptance.

- Performed troubleshooting, NMS activities, network upgrades, PAT/ATP and network health checks.
- Coordinated engineers, field teams, vendors and customers during network rollout.Worked on GPON MA5600T, U2000, MSAN and access-network service provisioning, testing and acceptance.
- Performed network health checks, NMS backup/restoration, vulnerability scanning and database access-control reviews.

MAJOR PROJECTS

- Saudi Electricity Company / National Grid – Optical network design, integration and support
- Saudi Telecom Company (STC) – Nokia optical transport deployment, upgrades and operations
- SEA-ME-WE 5 – 100G submarine cable network operations and maintenance
- Ciena upgradation Project
- 100G TW1 Submarine System & MSAN Upgrade – Huawei OTN

PROFESSIONAL AFFILIATIONS

- Pakistan Engineering Council — Electronics/Electrical Engineering registration
- Saudi Council of Engineers — Engineering registration

EDUCATION & CERTIFICATIONS

- PMP Certified — Project Management Institute (PMI) | October 2026
- CCNP Certified | Cisco | September 2026
- Nokia Services Expert | July 2026
- Nokia Optical professional | May 2026
- Nokia DWDM Project Training | July 2019
- Huawei Transmission Project Training | April 2019
- BS Electronics Engineering — Sir Syed University of Engineering & Technology, Karachi, Pakistan | 2008–2011 | CGPA 3.1
- Intermediate – Govt. College Formen , Karachi, Pakistan | 2006-2007 | A Grade
- High School – Placid High School, Karachi , Pakistan | 2005 | A1 Grade

LANGUAGES

- English — Professional
- Urdu — Native
- Arabic — Basic

ADDITIONAL INFORMATION

- References: Available upon request